

E131016_35

The equation $x^4 - x^2 + x + y^3 - y + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$x^4 - x^2 + x + y^3 - y + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = -5H_2$ and $d_2 = H_1$, and the product formula for the Hilbert symbol of one pair of values. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 - x^2 + x + y^3 - y + 3$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= 4x^2 - 23x + 3y - 25, \\ T_1 &= 123904x^8 - 3764992x^7 - 92928x^6y + 43218256x^6 + 2289408x^5y \\ &\quad - 216628568x^5 + 69696x^4y^2 - 19830396x^4y + 302680553x^4 \\ &\quad - 1316304x^3y^2 + 62755449x^3y + 934835149x^3 + 7739649x^2y^2 \\ &\quad + 9841170x^2y - 1618672249x^2 - 10790505xy^2 - 251330850xy - 3428716370x \\ &\quad - 21053475y^2 - 175445625y - 1435964785, \\ U_1 &= -704x^5 + 12720x^4 - 67019x^3 + 40836x^2 + 312998x + 189970, \\ V_1 &= 1737, \\ G_1 &= 209088x^4 - 3948912x^3 + 23218947x^2 - 32371515x - 63160425, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= x - y, \\ T_2 &= 4x^4 + 4x^3y - 4x^3 + 4x^2y^2 - 4x^2y - 12x^2 - 4xy^2 - 8xy + 4x \\ &\quad - 8y^2 + 5, \\ U_2 &= -2x^3 + 4x + 1, \\ V_2 &= 1, \\ G_2 &= -4x^2 + 4x + 8, \end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\begin{aligned} \lambda_1 &= 24320x^4 - 24320x^3y - 291408x^3 + 286848x^2y + 802940x^2 \\ &\quad + 18240xy^2 - 693296xy + 904868x - 110256y^2 - 918800y + 9207, \\ \mu_1 &= -97280x^5 + 1779712x^4 - 9636912x^3 + 7508228x^2 + 41234124x + 22666853, \\ \nu_1 &= -54720x + 330768, \end{aligned}$$

and, for the signs, we also use

$$\begin{aligned} Q_1 &= 16x^4 - 184x^3 - 12x^2y + 329x^2 + 69xy + 1150x + 9y^2 + 75y + 616, \\ M_1 &= 64x^6 - 1104x^5 + 5121x^4 + 1633x^3 - 32184x^2 - 42945x - 15481, \\ E_1 &= 432x^4 - 4968x^3 + 8883x^2 + 31050x + 16551, \end{aligned}$$

$$\begin{aligned}
Q_2 &= x^2 + xy + y^2 - 1, \\
M_2 &= x^4 + x^3 - x^2 + 3, \\
E_2 &= 3x^2 - 4.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 + 5H_2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - H_1V_2^2 = G_2F, \quad (3)$$

$$\lambda_1H_1 + \mu_1H_2 + \nu_1F = 3^4 \cdot 97^2, \quad (4)$$

$$27F = Q_1H_1 - M_1, \quad (5)$$

$$36Q_1 = (-12x^2 + 69x + 18y + 75)^2 + E_1. \quad (6)$$

$$-F = Q_2H_2 - M_2, \quad (7)$$

$$4Q_2 = (x + 2y)^2 + E_2. \quad (8)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$A = -d_2 = -H_1, \quad B = -d_1 = 5H_2, \quad T_A = -T_1, \quad T_B = 5T_2.$$

By (2)–(3), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$AT_A = U_1^2 - BV_1^2, \quad BT_B = (5U_2)^2 - A(5V_2)^2. \quad (9)$$

Step 1: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (5) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 9$, write $x = 9 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $-1 \leq x \leq 8$. For $x \in \{0, 2, 4, 6, 8\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x \in \{-1, 1, 5, 7\}$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with such x . For $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_1 > 0$ and $E_1 > 0$, so (6) gives $36Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (7) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-1 \leq x \leq 1$. For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -1$ and $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with such x . Hence $M_2 > 0$ and $E_2 > 0$, so (8) gives $4Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

By these signs, $A < 0$ and $B > 0$.

Hence Theorem 2(i) gives $(A, B)_\infty = 1$ (as $B > 0$).

Step 2: primes outside S . Let $S = \{2, 3, 5, 97, 193\}$ be the set of primes dividing 2, the constant 5 in the $d_k = cH_{k'}$, $V_1 = 1737$ and $V_2 = 1$, and the right-hand side of (4). Let $p \notin S$ be an odd prime; we show that $(A, B)_p = 1$. If $p \nmid AB$, this follows from Theorem 2(ii) with

$\alpha = \beta = 0$. If $p \mid A$, then $p \mid H_1$, so $p \nmid H_2$ by (4) and $p \nmid B$; as $p \nmid V_1$, Lemma 3 with $(a, b, r, c, t) = (A, B, U_1, V_1, T_A)$ and (9) gives $(A, B)_p = 1$; if $p \mid B$, then $p \mid H_2$, so $p \nmid H_1$ by (4) and $p \nmid A$; as $p \nmid 5V_2$, Lemma 3 with $(a, b, r, c, t) = (B, A, 5U_2, 5V_2, T_B)$ and (9) gives $(B, A)_p = 1$, which equals $(A, B)_p$ by Theorem 2(v).

Step 3: the primes in S . By Step 1, A and B are non-zero. If $T_A \neq 0$, then (9) and Lemma 4 give $(A, B)_v = (T_A, B)_v$, and if $T_B \neq 0$, they give $(A, B)_v = (B, A)_v = (T_B, A)_v = (A, T_B)_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 8^2 residue pairs modulo 8, we find that (1) has exactly 8 solutions modulo 8, and that at each of them one of the cases listed for $p = 2$ in Table 1 holds. Each case gives the stated value of $(A, B)_2$. Hence $(A, B)_2 = 1$.

The prime 3. Checking the 3^2 residue pairs modulo 3, we find that (1) has exactly 3 solutions modulo 3, and that at each of them one of the cases listed for $p = 3$ in Table 1 holds. Each case gives the stated value of $(A, B)_3$. Hence $(A, B)_3 = 1$.

The prime 5. Checking the 5^2 residue pairs modulo 25, we find that (1) has exactly 5 solutions modulo 25, and that at each of them one of the cases listed for $p = 5$ in Table 1 holds. Each case gives the stated value of $(A, B)_5$. Hence $(A, B)_5 = -1$.

The prime 97. Checking the 97^2 residue pairs modulo 97, we find that (1) has exactly 103 solutions modulo 97, and that at each of them one of the cases listed for $p = 97$ in Table 1 holds. Each case gives the stated value of $(A, B)_{97}$. Hence $(A, B)_{97} = 1$.

The prime 193. Checking the 193^2 residue pairs modulo 193, we find that (1) has exactly 194 solutions modulo 193, and that at each of them one of the cases listed for $p = 193$ in Table 1 holds. Each case gives the stated value of $(A, B)_{193}$. Hence $(A, B)_{193} = 1$.

Step 4: contradiction. By Theorem 2(iv), $(A, B)_\infty \prod_p (A, B)_p = 1$, the product over all primes, and by Step 2, the factors with $p \notin S$ are 1. Hence $\prod_{p \in S} (A, B)_p = (A, B)_\infty = 1$. But by Step 3, $\prod_{p \in S} (A, B)_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A, B)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 8				
1	(A, B)	any	$2^{2i} \cdot 1$	1
2	(A, T_B)	$2^{2i} \cdot 1$	$2^{2i} \cdot 1$	1
3	(A, T_B)	$2^{2i} \cdot 3$	$2^{2i} \cdot 1$	1
4	(A, T_B)	$2^{2i} \cdot 5$	$2^{2i} \cdot 1$	1
5	(A, T_B)	$2^{2i} \cdot 7$	$2^{2i} \cdot 1$	1

	symbol	first entry	second entry	value
$p = 3$, solutions modulo 3				
1	$3 \nmid AB$			1
2	(A, T_B)	$3^{2i}(+)$	$3^{2i}(+)$	1
$p = 5$, solutions modulo 25				
1	(A, B)	$5^{2i}(-)$	$5^{2i+1}(+)$	-1
$p = 97$, solutions modulo 97				
1	$97 \nmid AB$			1
2	(A, T_B)	$97^{2i}(+)$	$97^{2i}(-)$	1
3	(T_A, B)	$97^{2i}(+)$	any	1
$p = 193$, solutions modulo 193				
1	$193 \nmid AB$			1
2	(A, B)	any	$193^{2i}(+)$	1
3	(T_A, B)	$193^{2i}(+)$	$193^{2i}(+)$	1